Dynamic Arrest Follows Assembly on a Temperature-Scaled Common Clock in an Elastin-like Polypeptide

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Abstract

Small-angle X-ray photon correlation spectroscopy (SA-XPCS) records, on the same detector frames, how the structure of an assembling protein polymer grows and how

its motion stops. In the sequence-defined elastin-like polypeptide $(\mathrm{VPAVG})_{30}$, four loadings held between 28.00 and 28.75 °C all assemble, and their correlation functions decay in two steps whose fast amplitude falls as the sample arrests: at 28.50 °C it drops from 0.80 to 0.07 at 175 nm within an hour, and less far at every smaller scale down to 83 nm. One rescaling of the waiting time collapses all four temperatures onto a single master curve: arrest speeds up by a factor of e for every $T_s = 0.24 \pm 0.02$ K of heating. The low-Q SAXS intensity superposes with the same T_s. Assembly and arrest therefore run on one clock, with arrest following assembly, and every sample that arrests is half arrested at nearly the same low-Q intensity. A scaling temperature this sharp points to a rate-limiting step in which many contacts must close together.

Significance Statement

Proteins that assemble on heating can go on to stop moving, as when a condensate hardens or a gel sets, and the two are usually read from one measurement: a scattering pattern that has grown enough is taken as solid. We follow both at once in a designed protein polymer, recording X-ray speckle fifty thousand times per second on the frames that give its structure. Structure and motion run on one clock that speeds up by a factor of e for every quarter of a degree of heating. Arrest trails assembly and reaches the largest length scales first. One rescaling of time brings the arrest at every temperature onto one master curve.

Introduction

Many proteins assemble into dense phases that progressively lose their mobility, a process known as dynamic arrest. Biomolecular condensates, for example, become more viscous as they age,[1] and some convert from liquid droplets into solid aggregates, a transition linked to neurodegenerative diseases such as amyotrophic lateral sclerosis.[2,3] Heat-set protein gels such as cooked egg white, and protein scaffolds that set at body temperature, follow a similar

sequence: the proteins assemble first, and the network they form arrests later.[4–6] In each case, the properties that matter, from the viscosity of a condensate to the stiffness of a gel, depend on how far and how fast the assembly has arrested. Assembly and arrest are, however, usually followed with different tools. Scattering and microscopy record structure, whereas mobility is measured by microrheology, bulk rheology and, for condensates, fluorescence recovery after photobleaching, each at its own length scale.[7] Whether arrest simply tracks the growth of structure, and what sets its pace, have therefore been difficult to establish. The answers matter both for controlling when a protein material sets and for understanding how a condensate hardens.

Protein polymers whose assembly is encoded in their sequence are well suited to these questions, and elastin-like polypeptides (ELPs) are the best developed of them. ELPs are intrinsically disordered proteins whose sequence, molecular weight, and lower critical solution temperature (LCST) behavior can be programmed recombinantly,[8–12] and they have been proposed as minimal models of the disordered proteins that form condensates.[13,14] Above the LCST, ELPs undergo hydrophobic desolvation and can form coacervates or hierarchical assemblies depending on sequence, concentration, and thermal history.[11,12,15] In particular, alanine-containing ELPs can form stiff, thermoresponsive gels through arrested phase separation.[16,17] Although these studies establish the arrested state from time-averaged scattering and rheology, how an assembly evolves toward arrest, and what paces it, remains elusive.

Following an assembly toward arrest requires measuring its structure and its dynamics at the same time, because the two can evolve on different time scales. Small-angle X-ray photon correlation spectroscopy (SA-XPCS) measures both simultaneously, on the same detector frames: averaging the frames in time gives the SAXS profile, and correlating them gives the speckle decorrelation at the same low-Q length scales. Measuring them together places assembly and arrest in the same sample volume, at the same moment and on the same length scales, so the order and relative pace of the two can be read directly. In protein systems,

XPCS has resolved two relaxation modes with ballistic droplet motion during liquid–liquid phase separation,[18] temporally heterogeneous dynamics during gelation,[5] and stretched cage relaxation in crowded solutions.[19] A protein-rich morphology can likewise keep reorganizing through junction exchange, whose kinetics set the network relaxation time in associative networks,[20] and through the residual coarsening that follows arrested phase separation.[21,22] Both can proceed while the SAXS profile changes only modestly.

Here we use SA-XPCS to follow assembly and arrest together in concentrated aqueous $(\mathrm{VPAVG})_{30}$, an ELP built from 30 repeats of the pentapeptide valine-proline-alanine-valine-glycine. In isothermal experiments at 28.00, 28.25, 28.50 and 28.75 °C, the protein assembles at all four temperatures. The arrest that follows is strongly temperature dependent and sets in earlier at higher temperature. When the waiting time is rescaled by a single temperature-dependent factor, the arrest at all four temperatures falls onto one master curve. Arrest speeds up by a factor of e for every $T_s = 0.24 \pm 0.02$ K of heating, so warming by a quarter of a degree makes it almost three times faster. The growth of structure follows the same clock. This answers, for one protein polymer, the two questions raised at the outset: arrest trails assembly on a common clock, and temperature sets the pace of both. A temperature dependence this steep points to a rate-limiting step in which many contacts close together. For protein materials, the time to arrest thus becomes a quantity that can be measured directly and tuned by sub-degree changes in temperature. For condensates, whose disordered proteins phase separate much as ELPs do, a cooperative step that paces assembly and arrest alike may determine when they turn from liquid to solid.

Results

Fig. 1 summarizes the SA-XPCS experiment. All four loadings came from a single batch of $(\mathrm{VPAVG})_{30}$ at $101\,\mathrm{mg\,mL^{-1}}$, in the buffer given below. Each 2 s SA-XPCS acquisition at Advanced Photon Source (APS) beamline 8-ID-I (Materials and Methods) used a previously

unexposed sample position to limit radiation damage (SI Appendix, section 5); temporal and azimuthal averaging yielded $I(Q)$ and multi-tau correlation analysis yielded $g_2(Q, \Delta t)$.

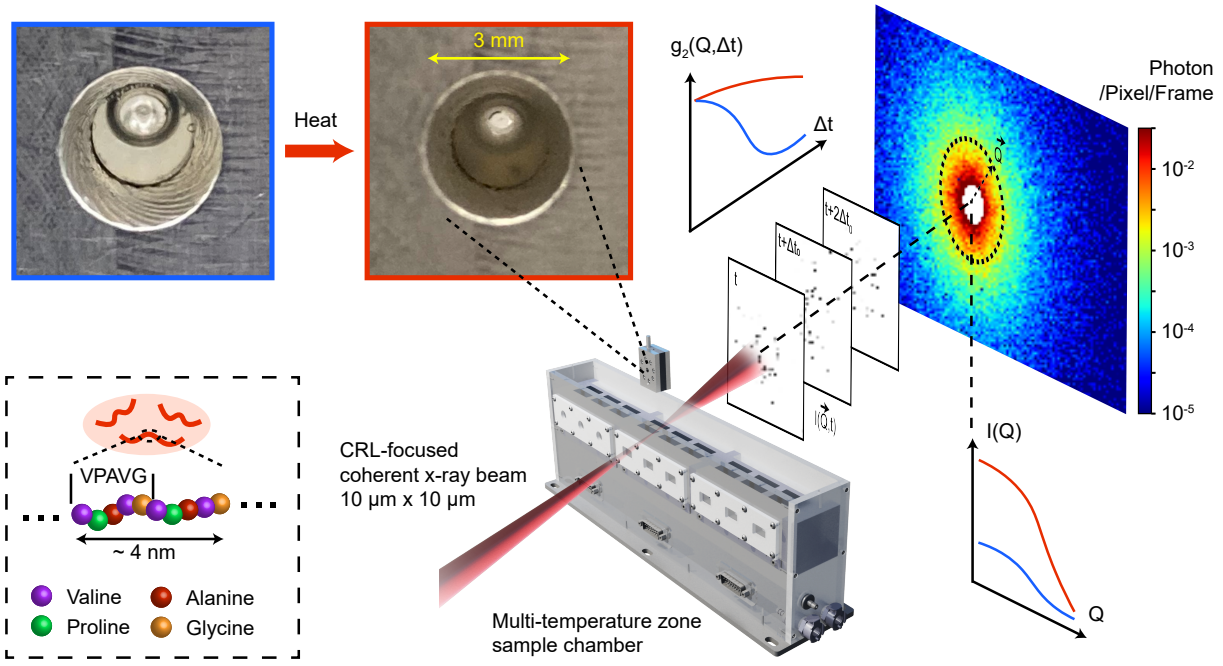


Figure 1: SA-XPCS measurement of thermally induced $(\mathrm{VPAVG})_{30}$ assembly at APS 8-ID-I. Each 100,000-frame set of sparse 20 µs frames feeds two analyses. Averaging them in time gives the SAXS intensity $I(Q)$: how much structure the sample has at each length scale $2\pi/Q$, which grows at low Q on heating. Correlating them in time gives $g_2(Q, \Delta t)$: how fast the structure at that same length scale rearranges. A decay in g_2 means the density fluctuations at $2\pi/Q$ are still moving; a flat g_2 means they are frozen over the accessible delay times Δt. Both are sketched beside the detector (red: assembled; blue: dispersed; illustrative, not data). Upper left: the aluminum cell, liquid chamber 3 mm across, photographed with the ELP sample below (blue outline) and above (red outline) the transition. Lower left: one $(\mathrm{VPAVG})_{30}$ repeat, residues colored as keyed. Right: the frame-averaged detector image, color bar in photons per pixel per frame, white center the beamstop mask; the graininess is sparse counting statistics, not resolved speckle.

To resolve mesoscale arrest isothermally, each of four loadings was equilibrated at 6 °C for at least 30 min and measured at that temperature. It was warmed from there to 20 °C at 1 °C min^{-1} and heated at 0.1 °C min^{-1} to a final temperature between 28.00 and 28.75 °C, where it was held. The first ramp ended at 20 °C because no appreciable assembly or dynamic arrest was observed at room temperature. The second ramp was kept slow to prevent the temperature from overshooting the target. The waiting time t_w starts at the first acquisition

with the calibrated resistance-temperature detector (RTD) at the cell reading the target. Fig. 2 shows the loading held at 28.50 °C; the other three are in SI Appendix, Figs. S4, S5 and S7, and Fig. 3 puts all four on one axis. Each t_w is the midpoint of a group of about 100 consecutive acquisitions at previously unexposed positions, averaged over about 10 min (SI Appendix, section 6.1). The SAXS intensity starts close to the 6 °C reference of the same loading and grows fourfold at low Q during the isothermal experiment, staying broad and monotonic with no peak emerging (Fig. 2a), so the assembly gains density contrast without settling on a single characteristic domain size. Correlation functions are fitted in the five lowest Q bins, 0.00359 to 0.00756 Å^{-1} or 175 to 83 nm; above them the scattered intensity has fallen by more than an order of magnitude and g_2 is dominated by counting noise.

The intensity autocorrelation function g_2 for a single SA-XPCS acquisition is defined as:

$$g_2(Q, \Delta t) = \frac{\langle I(Q,t) I(Q, t+\Delta t)\rangle_t}{\langle I(Q,t)\rangle_t^2}, \tag{1}$$

where $I(Q,t)$ is the coherent scattering intensity and Δt is the delay time. Through the Siegert relation, g_2 reports the relaxation of density fluctuations at length scale $2\pi/Q$. At the largest length scale the correlation functions show a two-step decay: a millisecond-scale relaxation followed by an apparent plateau that persists through the 2 s acquisition (Fig. 2b), the fast step being resolved only by the 50 kHz framing of the XSPA-500k.[23]

As t_w elapses over repeated acquisitions, the amplitude of the first decay decreases and the plateau rises. Such two-step relaxation and growth of a slow component are characteristic of heterogeneous kinetic arrest in protein and colloidal gels.[5,24,25] The two-step decays in g_2 were parameterized by:

$$\begin{aligned} g_2(Q, \Delta t, t_w) &= 1 + \beta \left| g_1(Q, \Delta t, t_w) \right|^2, \\ g_1(Q, \Delta t, t_w) &= f(Q, t_w) \exp\left[- \left(D_0(t_w)\, Q^2\, \Delta t \right)^{p_{\text{fast}}} \right] \\ &\quad + \left[1 - f(Q, t_w) \right] \exp\left[- \left(\frac{\Delta t}{\tau_{\text{slow}}(Q, t_w)} \right)^{p_{\text{slow}}(t_w)} \right], \end{aligned} \tag{2}$$

The first line is the Siegert relation, in which g_1 is the normalized intermediate scattering function and β is the instrumental speckle contrast, measured on a static reference and held fixed (SI Appendix, section 6.2). The second line is the two-mode model, where f is the fitted fast fraction and τ_{slow} an effective slow relaxation time. The effective fast relaxation time is replaced by $1/(D_0Q^2)$ in the fast term, where D_0 is a fitted diffusion coefficient. That form carries a physical claim: the population that is still mobile moves diffusively, so a density fluctuation of wavelength $2\pi/Q$ relaxes in the time a scatterer needs to diffuse across it and the relaxation rate grows as Q^2. Released from the constraint the fitted exponent has a median of -2.5, and the collapse of Fig. 3 survives releasing it (SI Appendix, section 4.2). The stretching exponents p_{fast} and p_{slow} describe the breadth of the relaxation-time distributions; both are shared across the five Q bins of a group and both are free from one group to the next. A stretched correlation function can still be diffusive as long as p_{fast} does not depend on Q and the characteristic time scales with Q^{-2}. The rate of the fast mode is reported as D_0, and its meaning does not depend on p_{fast}: at $\Delta t = 1/(D_0Q^2)$ the fast term has decayed to 1/e of its amplitude whatever the value of p_{fast}. D_0 is therefore the diffusion coefficient implied by the 1/e time of the fast decay, and it compares like with like between groups whose stretching differs. p_{fast} falls from about 1 while a sample is still mobile to about 0.2 once it has arrested, so a single value shared across an isothermal experiment is rejected by an F test at $p < 10^{-16}$ whenever the sample arrests. Holding p_{fast} at 1, which makes the fast mode a single exponential, describes the one sample that stays mobile but not the three that arrest, where the mean reduced χ^2 rises from 1.6 to between 7 and 10. The breadth of the fast relaxation is therefore a measured feature of arrest rather than a fitting convenience. A stretching exponent below 1 means the fast term is a distribution of relaxation times rather than a single rate, and the lower it falls the wider that distribution: the delay interval over which the fast term decays from 90 to 10% of its amplitude is $(\ln 10/\ln \frac{10}{9})^{1/p_{\mathrm{fast}}}$, which widens from 1.3 decades at $p_{\mathrm{fast}} = 1$ to 6.7 at 0.2. D_0 locates the fast decay on the delay axis, and p_{fast} sets how widely it is spread about that point. Late in an isothermal experiment, the

material that is still mobile relaxes across a very wide spread of time scales. τ_{slow} exceeds the acquisition window, so it is an extrapolated descriptor; the primary quantities are f and D_0.

Fig. 2c shows the fitted fast fraction falling at every measured Q during the isothermal experiment. At $t_w = 300\,\mathrm{s}$, f is 0.80 at $Q = 0.00359$ Å^{-1} (175 nm); by 3740 s that bin has fallen to 0.07, so the complementary arrested fraction $1 - f$ has gone from 0.20 to 0.93 in under an hour. At every analyzed time f also increases monotonically with Q, in all seven groups, so arrest develops first and most strongly at the largest length scale probed, 175 nm, and progressively less so toward 83 nm. That ordering holds within each group, so it reflects length scale rather than the trend with t_w, and it ties the growing low-Q SAXS structure to the arrest fraction measured on the very same frames. Note that XPCS does not locate the arrested regions, and the scattering contributions are weighted by structure and contrast, therefore $1 - f$ is not a geometric volume fraction.

The stars in Fig. 2c follow the rate of the fast decay. While the sample is still largely mobile, D_0 falls from 8.0 to 2.3×10^7 Å$^2\,$s^{-1}. Once f has dropped below 0.7 it stays between 0.9 and 1.2×10^7 Å$^2\,$s^{-1} while $1 - f$ climbs from 0.46 to 0.63, and the amplitude keeps falling, to 0.07, after the fast decay has become too weak to fix a rate (open stars). The longer isothermal experiment at 28.25 °C shows the same over a wider range: the rate stays within a factor of 1.7, with no trend, while f falls from 0.67 to 0.29 (SI Appendix, Fig. S8). Late in arrest, therefore, the amplitude of the fast decay keeps changing while its rate stays nearly constant. The fast relaxation also broadens, p_{fast} falling from 0.7–0.9 at the start to 0.20–0.35 at the end of each isothermal experiment in which the sample arrests. The same pairing of a plateauing fast time scale with a declining fast amplitude was found in a thermoreversible nanoparticle gel.[24]

The fast amplitude and the fast rate are not equally well determined late in an isothermal experiment: f is set by the plateau height and stays measured, whereas any fast-mode time scale needs a resolved decay and is lost once f falls to a few percent. τ_{slow} lies beyond the acquisition window and is not used for scaling. Fresh material at every acquisition keeps

cumulative dose from tracking t_w, so the trend in f cannot be a dose artifact, and a control at 22–41 times this flux shows no change in g_2 with dose rate (SI Appendix, Fig. S3).

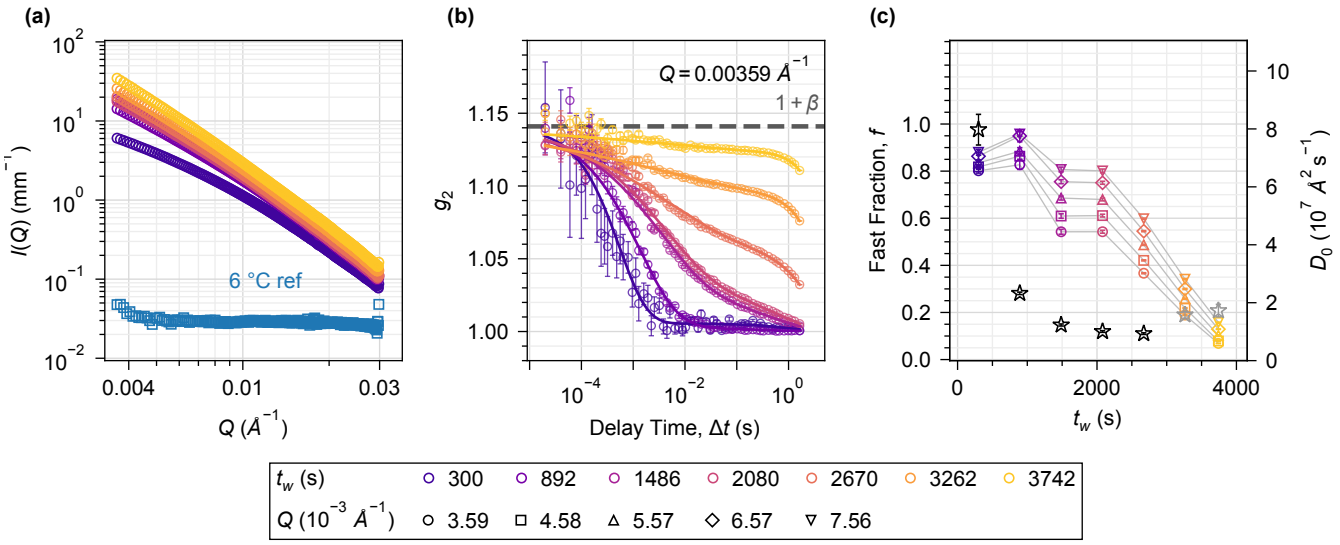


Figure 2: Assembly and dynamic arrest of $(\mathrm{VPAVG})_{30}$ held at 28.50 °C. During the isothermal experiment the mesoscale structure keeps growing while the dynamics stop: the low-Q intensity rises fourfold while the fast fraction falls from 0.80 to 0.07 at the largest length scale. (a) Background-subtracted SAXS on the absolute scale of SI Appendix, Fig. S2, with the empty cell subtracted as in SI Appendix, section 6.2 and the 6 °C reference of the same loading as blue squares; profiles are drawn from the lowest fitted Q bin outward. (b) g_2 at $Q = 0.00359$ Å^{-1} with fits of Eq. 2 at $\beta = 0.141$, p_{fast} free in each group; the dashed line is $1 + \beta$. (c) Left axis: the fast fraction f at the five Q values, marker shape denoting Q, grey lines guides to the eye. Right axis, stars: the fast-mode diffusion coefficient D_0, set by the 1/e time $1/(D_0 Q^2)$ of the fast decay and independent of p_{fast}; open grey stars mark groups with $f < 0.25$ at the lowest Q, where the fast decay is too weak to fix a rate. The rate falls threefold early in the experiment and then levels off while the amplitude keeps falling. The four remaining Q values are in SI Appendix, Fig. S6. Each t_w is the midpoint of a group averaging up to 100 acquisitions at fresh positions. Error bars are one-standard-deviation fit uncertainties; colour denotes waiting time, keyed along the bottom.

The other three loadings show what paces the arrest (Fig. 3; SI Appendix, section 6 and Figs. S4, S5 and S7). All four assemble, their low-Q intensity rising on the ramp and again during the isothermal experiment, but they arrest at very different rates: the time for f at the largest length scale to fall to 0.5 shortens 7.3-fold between 28.25 and 28.75 °C. One rescaling of the waiting time describes all four isothermal experiments. Against $X = t_w \exp[(T - T_0)/T_s]$, f at the largest length scale falls onto one master curve (Fig. 3b), with $T_s = 0.24 \pm 0.02$ K.

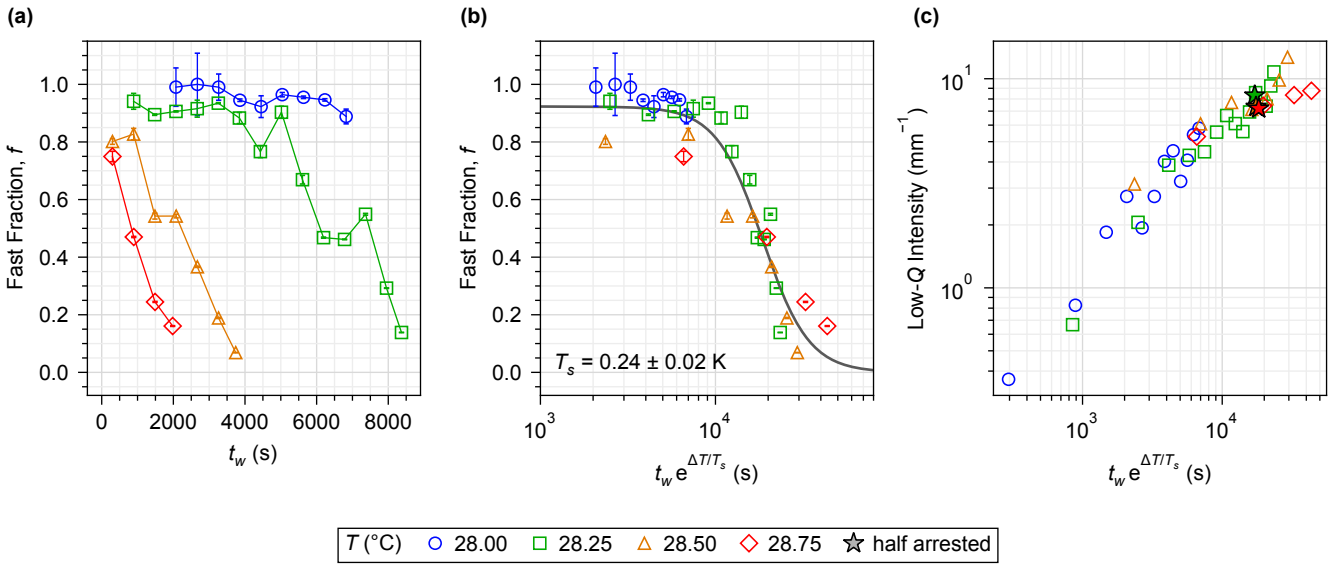


Figure 3: How fast arrest runs, and the single rescaling that collapses it at every temperature. (a) The fast fraction f at the largest length scale, $Q = 0.00359$ Å^{-1}, against t_w, the midpoint of each group. At 28.00 °C f stays at 0.89 or above for 6800 s; at the three higher temperatures it falls, and the fall begins earlier as the temperature rises. (b) The same f against $t_w\, e^{\Delta T/T_s}$, with $\Delta T = T - T_0$. The grey line is one master curve, $f = A/[1 + (X/X_c)^n]$, fitted to all four isothermal experiments with T_s free; it improves the residual threefold over no rescaling. $T_0 = 28.00$ °C is a convention that slides the curve without changing T_s, whose uncertainty is a bootstrap standard deviation over groups. The collapse is approximate: the curve lies about eight fit uncertainties from the points. (c) The low-Q intensity, $I(Q)$ averaged over 0.004 to 0.008 Å^{-1}, of every group of the four isothermal experiments against $t_w\, e^{\Delta T/T_s}$ with the T_s of (b). Stars mark where each sample that arrests is half arrested, $f = 0.5$ in (a). Colour and marker shape denote temperature, keyed along the bottom; error bars are one-standard-deviation fit uncertainties.

Collapsed one Q bin at a time, T_s reads 0.24 to 0.27 K, so the same temperature scale holds at every length scale measured, and the time scale of the master curve grows 1.3-fold from 175 to 83 nm (SI Appendix, Fig. S9). The collapse says more than that arrest is faster at higher temperature: one factor on the waiting time accounts for most of the difference between the four experiments, so temperature sets the pace of arrest. The 28.00 °C loading, which never arrests during its 6800 s isothermal experiment, is not an exception but the same curve seen too early: on the master curve it would be half arrested near 17 600 s.

The SAXS of the same frames runs on the same clock. Placed on $t_w \exp[(T - T_0)/T_s]$ with the T_s of the arrest, the low-Q intensity of all four isothermal experiments falls on one curve (Fig. 3c), and fitted on its own it returns the same $T_s = 0.24$ K. Assembly runs ahead of arrest along that clock. Each sample that arrests is half arrested when its low-Q intensity reaches 7.2 to 8.3 mm^{-1}, 175 to 220 times its level at 20 °C, and the intensity grows only 1.2- to 1.7-fold more while f falls from 0.5 to between 0.07 and 0.16. The isothermal experiment at 28.00 °C ends at 5.8 mm^{-1} with f still 0.89, short of that stage. Arrest is therefore a step that follows assembly, and most of it happens after most of the low-Q structure has formed, where the scattered intensity changes little and the arrested fraction is followed only by the dynamics.

A scaling temperature of a few tenths of a kelvin is itself the signature of a cooperative step. The same collapse was introduced for thermoreversible colloidal gels, where the formation rate varies with quench depth far more steeply than the pair attraction alone allows, and that steepness was attributed to the cooperative bonding needed to build locally favored structures.[26,27] Applied to that system it returns $T_s = 0.33$ and 0.45 K for two particle sizes, corresponding to local structures held by roughly twenty bonds.[24] A single contact cannot produce such steepness: one bond's energy changes far too little over half a degree. The temperature sensitivity of an individual contact here is not independently calibrated, so no bond number is extracted, but arrest being gated by something that must form many contacts at once is what the scaling requires.

Discussion

This changes what is understood to pace the assembly. Above the LCST, ELP assembly is driven by hydrophobic desolvation, and desolvation is well under way before any isothermal experiment begins: the scattered intensity has already grown two- to fourfold by 27.5 °C on the ramp (SI Appendix, section 6). What the scaling adds is that desolvation is not the step that sets the clock. The equilibrium of a single contact shifts very little over half a degree, yet half a degree changes the rate of arrest sevenfold, and the low-Q structure grows on the same clock. The rate-limiting event is therefore one that builds the mesoscale structure and then freezes it: the concerted formation of junctions in which many contacts close together.

The same steepness sets a practical scale. At $T_s = 0.24$ K, holding this sequence a quarter of a degree warmer shortens the time to arrest by a factor of 2.8, so the rate at which an ELP network assembles and arrests is adjustable over orders of magnitude by sub-degree temperature control.

The growth of low-Q SAXS intensity and the decline of the fast fraction are two aspects of network formation that run on one clock, with arrest following assembly. That sequence is consistent with network-forming arrested phase separation in a dynamically asymmetric protein–solvent mixture, where concentration fluctuations build a connected dense phase and viscoelastic constraints suppress relaxation.[5,18,21,28] Related ELPs form bicontinuous networks with stress-relaxation times beyond 10^3 s.[16,17] The cooperative step that the scaling points to is also familiar from protein systems, which clear a nucleation barrier before growth begins[29] and accelerate once started.[30]

What paces arrest is also an open question for biomolecular condensates, whose maturation from liquid to solid governs their function and their role in aggregation.[31,32] Condensates are too small and too dilute for the measurement reported here, but a concentrated, sequence-defined mimic is not. In this mimic, arrest follows assembly on a common clock, and the steep temperature dependence of that clock points to a cooperative step. Whether condensates harden through a step of the same kind can be tested with the tools already used on them: if

they do, their arrest followed at a few temperatures should collapse under one rescaling of time.

Two limitations bound the interpretation, and each points to the measurement that would lift it. First, the 2×10^{-5}–1.6 s delay range quantifies the fraction arrested on that scale rather than the infinite-time nonergodicity factor, for which linear shear rheology on a scaled-up preparation is the complement. Second, the scaling rests on three arresting temperatures spanning half a degree, and the collapse is approximate: rescaling improves the residual threefold but not to the measurement uncertainty. More temperatures would settle it. T_0 is a convention rather than a measured transition value, so no reference time is quoted from it.

Subject to those constraints, the measurements give one picture. SA-XPCS follows what SAXS alone cannot: how far an assembly has arrested. Structure and arrest run on one clock, with arrest following assembly and finishing after most of the low-Q structure has formed. Arrest reaches the largest length scales first. One rescaling of time brings every isothermal experiment onto one master curve, so temperature sets the pace of arrest. That clock runs almost three times faster for every quarter of a degree of heating, a steepness that points to a rate-limiting step in which many contacts close together.

Materials and Methods

The Sanger-verified construct encoded an N-terminal His-tagged $(\mathrm{VPAVG})_{30}$ protein (complete sequence in SI Appendix). Recombinant $(\mathrm{VPAVG})_{30}$ was expressed in *E. coli*, purified by immobilized metal-affinity chromatography, and buffer exchanged into 20 mmol L^{-1} HEPES, 250 mmol L^{-1} NaCl, and 2 mmol L^{-1} DTT at pH 8. Expression, solubility, and purity were monitored by SDS-PAGE (SI Appendix, Fig. S1b). Protein concentration was determined from A_{280} on a NanoDrop spectrophotometer with $E_{280}^{1\%} = 0.95$; the batch used for all four loadings was at 101 mg mL^{-1}.

SA-XPCS was performed at APS 8-ID-I with a 10.94 keV, 10×10 $\mu\mathrm{m}^2$ partially coherent

beam and a Rigaku XSPA-500k detector 7.8 m downstream. Each acquisition contained 100,000 frames at 50 kHz (20 µs per frame, 2 s total), and the four isothermal experiments were carried out in one beamtime in a single cell block. Further acquisition, fitting and radiation-control details are in SI Appendix.

The four loadings were measured one after another on the same temperature zone and resistance-temperature detector, in the order 28.00, 28.50, 28.25 and 28.75 °C, and the detector reading varied by 0.004 °C (standard deviation) during each isothermal experiment. Acquisitions followed one another every 6 s, each at a fresh position. Consecutive acquisitions were averaged in groups of about 100 after two outlier cuts (SI Appendix, section 5.3), and each group is placed at the midpoint of the acquisitions it contains, $t_w = 0$ being the first acquisition with the detector at its target. Eq. 2 was fitted to the five lowest Q bins of each group at once by error-weighted least squares, with β fixed at the value measured on a static reference, f and τ_{slow} free in each bin, and D_0, p_{fast} and p_{slow} shared across the bins of a group (SI Appendix, section 4).

For the scaling analysis, f at the lowest Q of every group of the four isothermal experiments was fitted with one master curve, $f = A/[1+(X/X_c)^n]$ with $X = t_w \exp[(T-T_0)/T_s]$, weighting each point by its fit uncertainty floored at 0.01, and each higher Q bin was collapsed the same way on its own. $T_0 = 28.00$ °C is a convention: it moves the curve along X without changing T_s. The uncertainty of T_s is the standard deviation over bootstrap resamples of whole groups. The low-Q intensity is the mean absolute $I(Q)$ of each group over 0.004 to 0.008 Å^{-1} (SI Appendix, section 6.5).

Data, Materials, and Software Availability

Reduced data and analysis/figure-generation code are publicly available at `https://github.com/qzhang234/PubCode-2021-IDPP` [**archived at Zenodo; DOI to be added**]. The repository includes the 8-ID-I datasets used for the figures, radiation-control data, and analysis

scripts, and every data-derived figure can be regenerated from it alone. Raw beamline data are available from the corresponding authors upon reasonable request.

Author Contributions

[**To be completed by the authors.**]

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AI-assisted manuscript and code preparation. The authors used Anthropic's Claude models, accessed through Argo, Argonne National Laboratory's internal generative AI interface, to assist with manuscript preparation and with development of the data-analysis and figure-generation code (see SI Appendix). That code and the reduced data it operates on are publicly available at `https://github.com/qzhang234/PubCode-2021-IDPP`, so the reported results can be regenerated and checked independently. The authors take full responsibility for all manuscript content and reported results.

Competing Interest Statement

The authors declare no competing interest.

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